Complete UQFF Equations Reference

Version: 4.5.0 **Date:** May 2026 **Status:** Master Reference Document

Overview

This document provides a complete, canonical reference for all UQFF (Unified Quantum Field Framework) equations, constants, and derived relationships used throughout the Star-Magic research archive.

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Fundamental Constants

Physical Constants

Constant	Symbol	Value	Unit
Speed of Light	c	2.998×10^8	m/s
Planck Constant	$\hbar$	1.055×10^{-34}	$\mathrm{J \cdot s}$
Gravitational Constant	G	6.674×10^{-11}	$\mathrm{m^3/kg \cdot s^2}$
Solar Mass	$M_\odot$	1.989×10^{30}	kg
Solar Radius	$R_\odot$	6.96×10^8	m

UQFF-Specific Constants

Constant	Symbol	Value	Unit
SCm Vacuum Density	ρ_A	7.09×10^{-37}	$\mathrm{J/m^3}$
UA Vacuum Density	ρ_{UA}	7.09×10^{-36}	$\mathrm{J/m^3}$
SCm Velocity	V_{SCm}	$c/3 = 10^8$	m/s
Buoyancy Parameter	β_i	0.603	dimensionless
Vacuum Decay Rate	κ	0.0005	/day
String Resonance	$[SS_q]$	0.57	dimensionless
SCm Heaviside Factor	H_{SCm}	0.99	dimensionless
UA Suppression	U_{UA}	0.0001	dimensionless

Master UQFF Equations

Complete Unified Field (F__U)

The complete unified field is expressed as:

$$F_U = \sum_{i=1}^{26} [Ug1_i + Ug2_i + Ug3_i + Ug4_i] - Ubi + Um$$

This represents the total gravitational and quantum field force.

Decomposition by Components

- **Ug1-4:** Gravitational force components (26 layers each)
- **Ubi:** Buoyancy force correction
- **Um:** Universal magnetism term

Ug Terms

Ug1: Magnetic Dipole Force (26-Layer)

$$Ug1_i = k_1 \cdot \mu_s^{(i)} \cdot \frac{M}{r_i^2} \cdot e^{-\alpha t} \cdot \cos(\pi t_n) \cdot (1 + \delta_{def})$$

where: - k_1 is the coupling constant - $\mu_s^{(i)} = \rho_A \cdot V_i$ is the magnetic moment of layer i - α is the decay rate - δ_{def} is the deformation factor

Ug2: Charge-Reactivity Coupling

$$Ug2_i = k_2 \cdot (Q_{SCm} + Q_{UA}) \cdot \frac{M}{r_i^2} \cdot S(r - R_b) \cdot (1 + \delta_{sw} v_{sw}) \cdot H_{SCm} \cdot E_{react}$$

where: - Q_{SCm}, Q_{UA} are charge densities - $S(r - R_b)$ is the Heaviside step function (boundary at heliopause) - v_{sw} is the solar wind velocity - E_{react} is the reaction energy

Ug3: Magnetic String Rotation

$$Ug3_i = k_3 \cdot B_{disk} \cdot \cos(\omega_s t \pi) \cdot P_{core} \cdot E_{react}$$

where: - B_{disk} is the disk magnetic field - ω_s is the angular velocity - P_{core} is the core pressure

Ug4: Vacuum Concentration

$$Ug4_i = k_4 \cdot \rho_{vac} \cdot C_{concentration} \cdot e^{-\alpha t} \cdot \cos(\pi t_n)$$

where: - ρ_{vac} is the vacuum energy density - $C_{concentration}$ is the concentration factor

Buoyancy and Magnetism

Buoyancy Force (Ubi)

$$Ubi = \beta_i \cdot Ug_i \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot (1 + \epsilon_{sw}\rho_{sw}) \cdot \rho_A \cdot \cos(\pi t_n)$$

where: - $\beta_i \approx 0.603$ is the buoyancy coefficient - Ω_g is the galactic angular velocity - M_{bh} is the black hole mass - d_g is the galactic distance

Universal Magnetism (Um)

$$Um = \frac{\mu}{r^3}$$

where $\mu = M \cdot R^2 \cdot \omega_0$ is the magnetic moment.

26-Layer Decomposition

The UQFF framework decomposes the unified field into 26 independent layers:

$$F_U = \sum_{i=1}^{26} F_i^{(26)}$$

Each layer i contains contributions from all Ug terms with layer-specific parameters: - Radius scaling: $r_i = r/i$ - Density scaling: Depends on layer composition - Frequency scaling: $\omega_i \sim i$

MUGE Framework

Master Universal Gravity Equation (MUGE)

MUGE represents gravity as frequency-resonance driven:

$$g(r,t) = \sum_j g_j(r,t)$$

where each component is derived from fundamental resonance modes.

MUGE Compressed Form

$$g_{comp} = F_{DPM} \cdot f_{DPM} \cdot \frac{E_{vac,neb}}{c \cdot V_{sys}} + \text{correction terms}$$

DPM Base: Di-Pseudo-Monopole

$$F_{DPM} = I \cdot A \cdot (\omega_1 - \omega_2)$$

where: - I is current - A is area - ω_1, ω_2 are frequency differences

Source4 Integration

SOURCE4 Functions

SOURCE4 provides 37 validated physics functions:

UQFF Set (8 functions): - compute_FU_SOURCE4: Complete unified field - compute_Ug1_SOURCE4: Magnetic dipole - compute_Ug2_SOURCE4: Charge-reactivity - compute_Ug3_SOURCE4: String rotation - compute_Ug4_SOURCE4: Vacuum concentration - compute_Ubi_SOURCE4: Buoyancy - compute_Um_SOURCE4: Magnetism

MUGE Compressed (10 functions): Resonance-corrected gravity **MUGE Resonance (14 functions):** Full resonance modes **Helper Functions (6 functions):** Supporting calculations

Pre-defined Systems

Seven astrophysical systems with validated parameters: - SGR1745: Magnetar - SagA: Supermassive black hole - Tapestry: Star formation region - Westerlund2: Star cluster - Pillars: Star formation - Rings: Gravitational lens - Student Universe: Cosmological test case

Validation Status

- **Solvability:** 99.9% verified (Grok 4, Sept 2025)
- **Calibration:** Complete (, [SSq], H_SCm, U_UA, _i)
- **Cross-validation:** UQFF vs MUGE Compressed vs MUGE Resonance
- **Papers:** 935+ whitepapers documenting applications

References

All equations appear in the PAPER_001-935 archive.

Key papers: - Canonical Ug derivations: PAPER_XXX - MUGE framework: PAPER_YYY - SOURCE4 integration: PAPER_ZZZ

Revision History

Version	Date	Changes
4.5.0	May 2026	Complete reference including Phase H
4.4.0	Mar 2026	SOURCE4 integration
4.3.0	Feb 2026	Grok thread batch integrations
4.2.0	Jan 2026	MUGE compression framework
4.1.0	Dec 2025	SOURCE4 unified field
4.0.0	Sept 2025	99.9% solvability verification

This document is the canonical reference for all UQFF equations. All calculations in the Star-Magic archive reference these definitions.